

E0037

The equation $2y^3 + xy + x^4 + 1 = 0$ has no integer solutions

Size $h = 37$, length $l = 10$

Theorem 1. *The equation*

$$2y^3 + xy + x^4 + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $HT = U^2 + V^2$ on the curve (1), and the fact that a prime p with $p \equiv 3 \pmod{4}$ divides $U^2 + V^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only Fermat's little theorem.

1 The norm form $a^2 + b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N .

Lemma 2. *Let p be a prime with $p \equiv 3 \pmod{4}$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$ and put $r = ac$. Then $r^2 \equiv -(bc)^2 \equiv -1 \pmod{p}$, so $p \nmid r$, and Fermat's little theorem gives

$$1 \equiv r^{p-1} = (r^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2} = -1 \pmod{p},$$

because $(p-1)/2$ is odd. Then $p \mid 2$, which is impossible. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let p be a prime with $p \equiv 3 \pmod{4}$, and let $a, b \in \mathbb{Z}$ be not both zero. Then $v_p(a^2 + b^2)$ is even.*

Proof. Let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + b'^2$ by Lemma 2. Since $a^2 + b^2 = p^{2k}(a'^2 + b'^2)$, we get $v_p(a^2 + b^2) = 2k$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = 2y^3 + xy + x^4 + 1$, so that (1) is the equation $F(x, y) = 0$. Define

$$\begin{aligned} H_1 &= 2y^2 + x, & T_1 &= -y, & U_1 &= x^2, & V_1 &= 1, & G_1 &= -1, \\ H_2 &= x - y, & T_2 &= 2y^2 + 2xy + 2x^2 + x, & U_2 &= x^2 + x, & V_2 &= 1, & G_2 &= -1. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_k T_k - U_k^2 - V_k^2 = G_k F \quad (k = 1, 2), \tag{2}$$

$$T_2 = y^2 + (x + y)^2 + x(x + 1). \tag{3}$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) gives

$$H_1 T_1 = U_1^2 + V_1^2, \quad H_2 T_2 = U_2^2 + V_2^2, \quad (4)$$

where H_k, T_k, U_k and V_k now denote the values of these polynomials at (x, y) .

Step 1: congruences. Checking the 16 residue pairs modulo 4, we find that the only solutions of (1) modulo 4 are $(x, y) \equiv (1, 2)$ and $(x, y) \equiv (3, 2)$. In these cases $(H_1, H_2) \equiv (1, 3)$ and $(H_1, H_2) \equiv (3, 1) \pmod{4}$, respectively. In particular, $y \neq 0$, H_1 and H_2 are odd, and one of them is congruent to 3 modulo 4.

Step 2: signs. Since $x(x+1) \geq 0$ for every integer x and $y \neq 0$, (3) gives $T_2 \geq y^2 > 0$, and (4) gives $H_2 > 0$, that is, $x > y$. If $y > 0$, then $x > y > 0$, and every term on the left-hand side of (1) is positive, which is impossible. Hence $y < 0$, so $T_1 = -y > 0$, and (4) gives $H_1 > 0$.

Step 3: primes with $p \equiv 3 \pmod{4}$. Let p be a prime with $p \equiv 3 \pmod{4}$, and let $k \in \{1, 2\}$. We show that $v_p(H_k)$ is even. Since $V_k = 1$, (4) and Lemma 3 show that $v_p(H_k T_k) = v_p(U_k^2 + V_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + V_k^2$ by (4), so $p \mid V_k$ by Lemma 2, which is impossible because $V_k = 1$. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_k T_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Steps 1 and 2, H_1 and H_2 are positive odd integers. Every prime factor p of H_k with $p \not\equiv 3 \pmod{4}$ satisfies $p \equiv 1 \pmod{4}$, and every prime factor with $p \equiv 3 \pmod{4}$ occurs with an even exponent by Step 3. Since the residue 1 modulo 4 is closed under multiplication and contains the squares of all odd integers, $H_1 \equiv H_2 \equiv 1 \pmod{4}$. This contradicts Step 1, and the theorem is proved. $\square$